

1)

If q_1, q_2, q_3 , and q_4 are the four angles then

(a)

$$H_{01} = \begin{bmatrix} \cos(q_1) & -\sin(q_1) & 0 & 0 \\ \sin(q_1) & \cos(q_1) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_{12} = \begin{bmatrix} \cos(q_2) & 0 & \sin(q_2) & 0 \\ 0 & 1 & 0 & 0 \\ -\sin(q_2) & 0 & \cos(q_2) & 0.077 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_{23} = \begin{bmatrix} \cos(q_3) & 0 & \sin(q_3) & 0.024 \\ 0 & 1 & 0 & 0 \\ -\sin(q_3) & 0 & \cos(q_3) & 0.128 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$H_{34} = \begin{bmatrix} \cos(q_4) & 0 & \sin(q_4) & 0.124 \\ 0 & 1 & 0 & 0 \\ -\sin(q_4) & 0 & \cos(q_4) & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

(b)

$$E_1 = [0.126 \ 0 \ 0 \ 1]'$$

$$E_0 = H_{01} * H_{12} * H_{23} * H_{34} * E_1$$

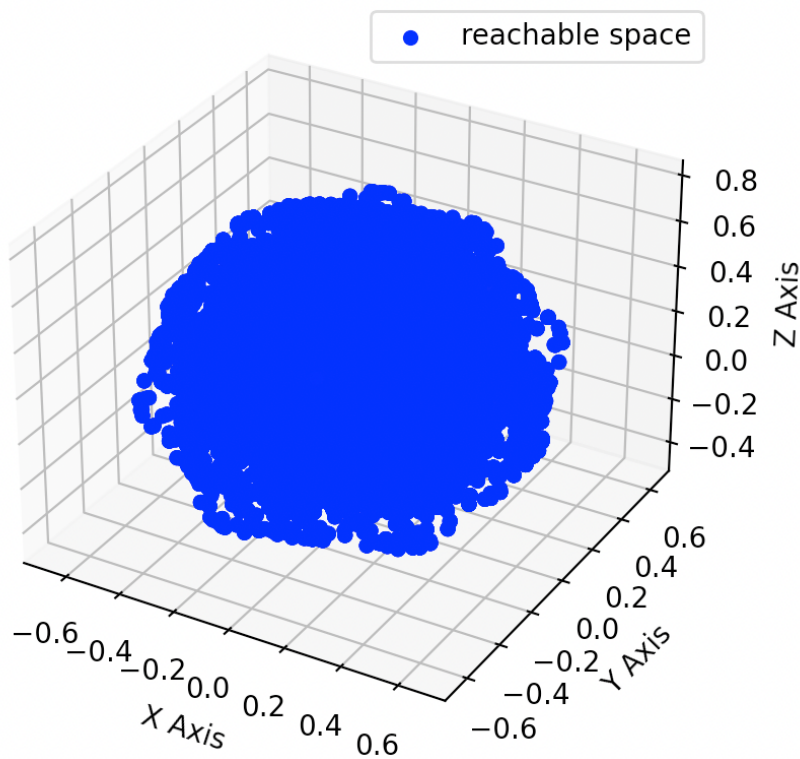
2

(a)

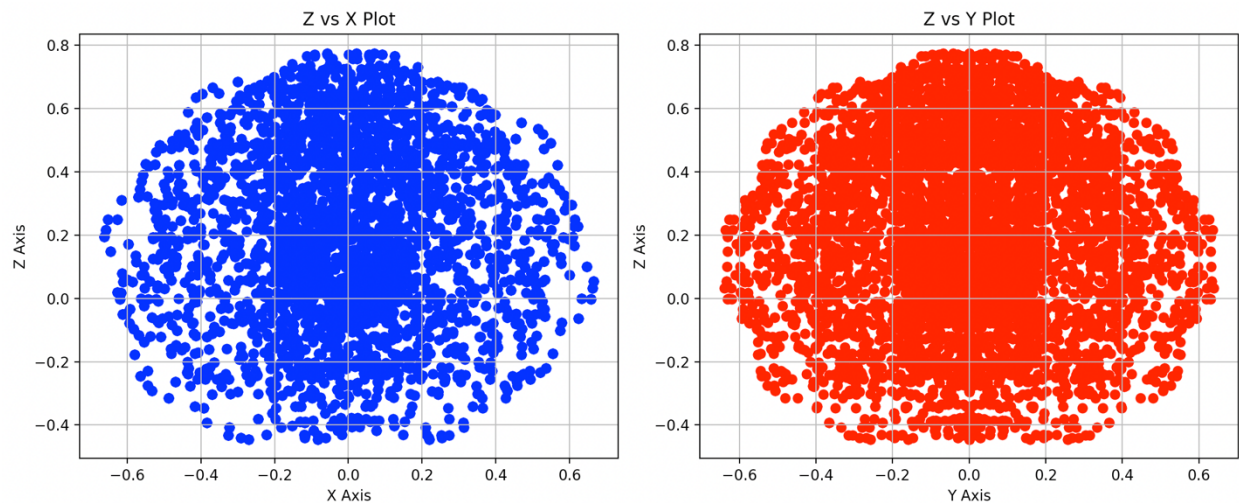
For the joint angles: $q = \text{np.array}([0.5123, -1.6532, 0.5439, -0.3432, -0.1125, 0.2342]);$ Position of the end-effector = $[-0.0767, -0.0360, 0.5181]$ Orientation of the end-effector (quaternion) = $[0.7962, 0.0917, -0.5661, 0.1930]$

(b)

3D Plot of Reachable Space



Here is a 2D plot of the reachable space: z vs x and z vs y.



(c)

Here is some pseudo-code

Given: position_ref, quaternion_ref

Min_distance = 1000 #some large value

For q1 in range

For q2 in range

...

For q6 in range

q_test = q1(i),q2(j),...q6

position,orientation = forward_kinematics(q_test)

#compute euclidean distance, tune a weight w if needed

Distance = (position-position_ref)^2+w*(orientation- orientation_ref)^2

If (Distance<min_distance)

Save q_test